

CS101

if/else Statements

2/10

Spot 3 Differences

```
public class Calculator {  
    public static void main(String[] args) {  
        int balance = 1000;  
        int deposit = 250;  
        int withdrawal = 75;  
  
        // Update balance  
        balance += deposit;  
        balance -= withdrawal;  
  
        // Calculate fee (2% of balance)  
        double fee = balance * 0.02;  
  
        // Check if divisible by 100  
        int remainder = balance % 100;  
  
        System.out.println("Balance: $" + balance);  
        System.out.println("Fee: $" + fee);  
    }  
}
```

```
public class Calculator {  
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        int balance = 1000;  
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        int withdrawal = 75;  
  
        // Update balance  
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        // Calculate fee (2% of balance)  
        double fee = balance x 0.02;  
  
        // Check if divisible by 100  
        int remainder = balance % 100;  
  
        System.out.println("Balance: $" + balance);  
        System.out.println("Fee: $" + fee);  
    }  
}
```

Agenda

1. Demo, programmerVernacular
2. Alex Lee + w3 Tutorials (as group)
3. Work time
4. Quiz4 Practice (Kahoot)
5. Touch Typing

Announcements

1. No Class 2/17
2. No Quiz 2/19
3. Extended labs to Friday nights
4. CS101 GitHub [README.md](#) reversed Topic order

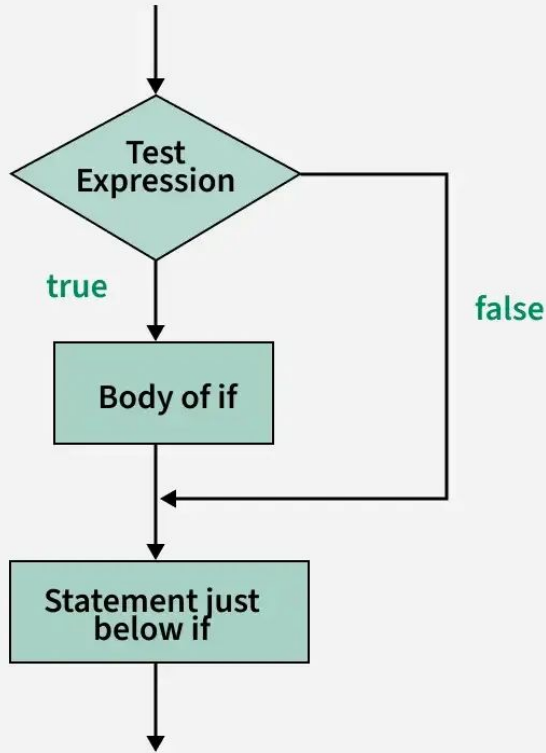
Alex Lee Boolean Tutorial

<https://www.youtube.com/watch?v=CHVVEGRGiJU>

W3 if/else Tutorial

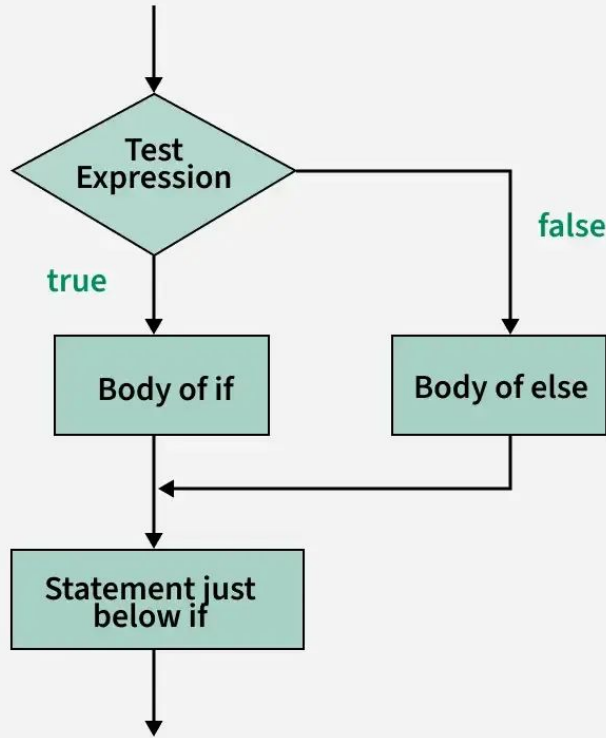
https://www.w3schools.com/java/java_conditions.asp

Control Flow Diagram: “if” statement



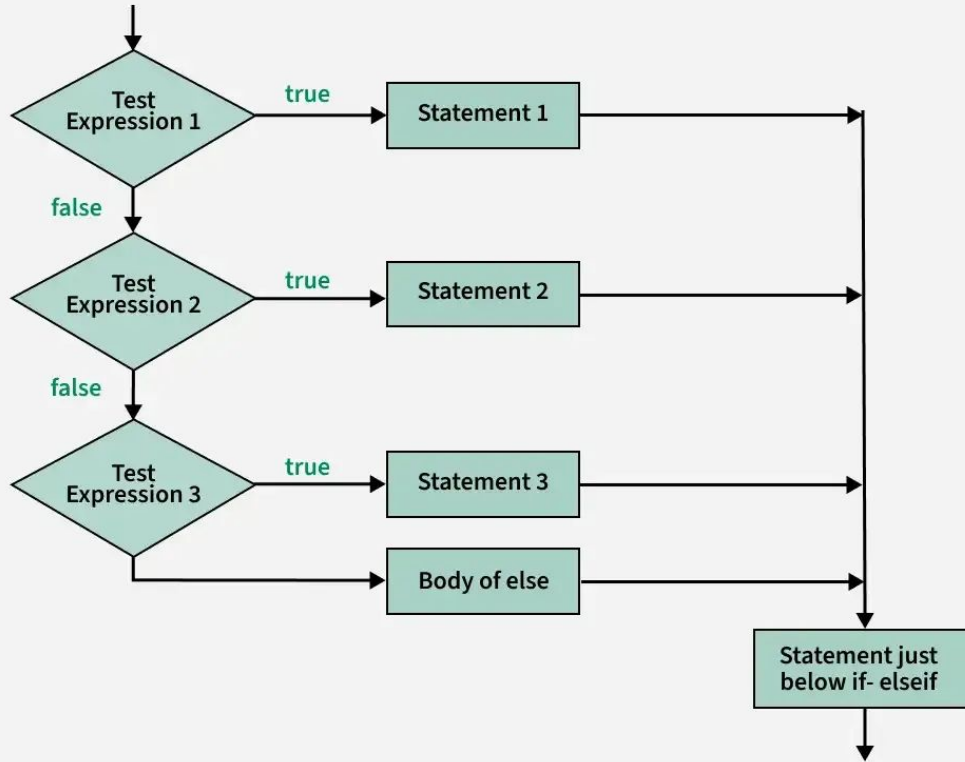
```
int x = 4;  
if (x > 10) {  
    // Run this code if `true`  
}
```

Control Flow Diagram: “if/else” statement



```
int x = 4;  
if (x > 10) {  
    // Run this code if `true`  
} else {  
    // Run this code if `false`  
}
```


Control Flow Diagram: “if/elseif” ladder



```
int x = 4;

if (x > 10) {
    // Run this code if `true`
    System.out.println("x > 10");
} else if (x > 5){
    // Run if first condition false, but this is true
    System.out.println("x > 5 and x <= 10");
} else if (x > 0){
    // Run if both above false, but this is true
    System.out.println("x > 0 and x <= 5");
} else {
    // Run if all above false
    System.out.println("x <= 0");
}
```

Before Next Class:

- Start HW2! Short demo and time for Qs next class
- Submit Lab1 and Lab2, due FRIDAY!!
- Due dates: canvas.arcadia.edu:

Next Class 2/12

1. Lab5, Booleans and Conditionals

- **Bring Headphones every class!**